Geospatial Time Machine: A Generative Model to Enhance Spectral–Temporal Data Resolution

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***Abstract*— Geospatial artificial intelligence (GeoAI) and data processing techniques have significantly advanced object detection, prediction, and classification tasks. However, the availability of machine learning-ready, labeled data for specific applications such as plant disease detection remains the major challenge for the broader adoption of GeoAI. For instance, collecting temporal unmanned aerial vehicle (UAV) imagery of agricultural crops to track disease emergence and progress requires substantial human labor and resources, which is often limited to a small spatial scale. Recognizing the pivotal role of temporal data in pattern recognition, object detection, and scene reconstruction, we introduce an innovative approach to augment multispectral temporal datasets: the geospatial time machine (GTM). Our proposed methodology combines graph neural network (GNN) and generative adversarial network (GAN) architectures to generate comprehensive synthetic temporal data encompassing multivariate time series. The results demonstrate that imagery generated through backcasting can enhance the accuracy of downstream classification tasks by up to 53% in plant disease detection, particularly in the initial stages of analyzing a crop growth using multispectral and multitemporal datasets.**

***Index Terms*— Data augmentation, deep learning, generative adversarial networks (GANs), generative artificial intelligence (GenAI), graph neural networks (GNNs), multispectral.**

I. Introduction

In RECENT years, data imputation and augmentation techniques have witnessed substantial advancements, revolutionizing object detection, prediction, and classification domains. Despite the ever-expanding volume of available data, a persistent challenge continues to be the lack of labeled imagery data for various applications. It is especially a prominent issue in the remote sensing of agriculture, where collecting high temporal data from unmanned aerial vehicle (UAV) imagery of crops at various growth stages is expensive. Satellite data provide frequent observations, but the spatial resolution is coarse, hampering plot-level phenotyping or tracking the emergence of plant disease.

Received 31 December 2024; accepted 4 March 2025. Date of publication 18 March 2025; date of current version 17 April 2025. This work was supported in part by NSF under Award 2133407, in part by NSF/United States Department of Agriculture (USDA) under Grant 2020-67021-31530, and in part by the USGS AmericaView under Grant G18AP0007. *(Corresponding author: Vasit Sagan.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3552629, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3552629

To cope with such challenges, considerable effort has been devoted to enhancing data augmentation techniques [1]. However, spatiotemporal imagery datasets paired with real-world application scenarios are lacking. A growing body of evidence highlights the significance of more comprehensive temporal data for improved yield prediction (see [2], [3]), anomaly detection [4], or classification tasks [5]. A similar significance is observed for multispectral data [6], when a higher number of samples usually leads to higher accuracy when training deep learning models [7]. In addition, the majority of the literature tends to focus more on predictive forecasting (e.g., land-cover and land-use [8]), and data fusion [9], [10], [11]; however, the historical temporal data plays a pivotal role in enhancing the accuracy of prediction models [12]. This importance was also acknowledged by Shen et al. [13], which presented a comprehensive review of scene reconstruction techniques, highlighting the significance of temporal-based techniques when reconstructing missing information and performing prediction or classification tasks.

Current endeavors in generative artificial intelligence (GenAI) to tackle scene and time-series reconstruction challenges are still in the early stage, encountering hurdles, such as ensuring adherence to physics constraints within the distribution [14] or generating probabilistic synthetic data [15] rather than linear synthetic samples. In remote sensing domains, where there is a high cost associated with real-time monitoring (e.g., multiple UAV flights for monitoring crops), there is a clear opportunity for using GenAI to develop and apply more robust synthetic scene reconstruction methods, especially for multispectral data. While there are several initiatives to generate synthetic time series (e.g., TimeGAN [16] and GT-generative adversarial network (GAN) [17]) and to perform scene reconstruction of RGB imagery [18], [19], there is a gap in methods to generate synthetic time series for multispectral datasets.

Current research efforts address the issue of temporal scarcity in multispectral data by employing imagery fusion techniques [10], [11], [20], which are also costly and prone to calibration errors [21]. Observing the lack of a multitemporal physics-based generative model for the multispectral context, this article aims to address the question of whether a GenAI-based method can generate synthetic temporal multispectral data that accurately mimics real imagery from past temporal records of a designated study area.

To that end, we introduce the geospatial time machine (GTM) concept, an innovative approach to augment

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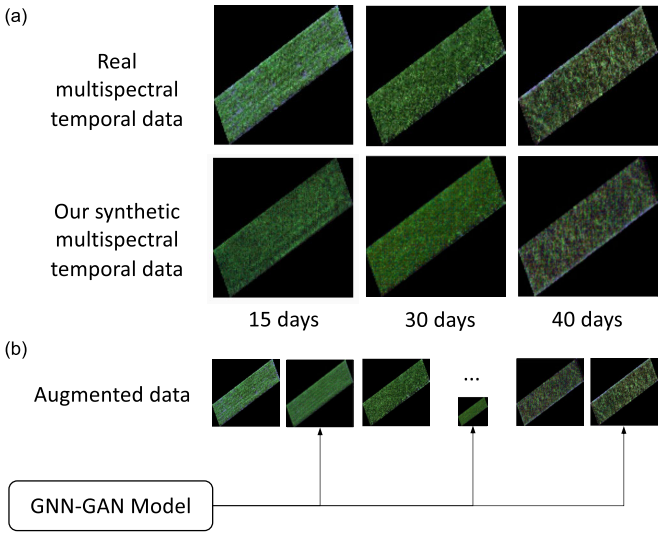


Fig. 1. (a) Comparison between real imagery of temporal plant growth data and synthetic samples generated from our GNN–GAN model. (b) Illustration shows the possible outputs from generating samples for other instances of datasets with limited temporal data.

multispectral and temporal datasets through a synthetic reconstruction of nonobserved time instants and their imagery. This approach involves synthetic data generation based on multivariate time series, opening up new avenues for improved analysis and prediction models utilizing synthetically reconstructed data.

GTM combines a graph neural network (GNN) with a GAN to accomplish the following: 1) capture the profiles of feature variation (e.g., vegetation indices and spectra) according to temporal snapshots; 2) utilize such variation profiles as discriminators for a multispectral GAN model; and 3) generate novel synthetic multispectral imagery corresponding to a specified time instant.

We first utilize image segmentation to convert multispectral images into graphs to obtain the profile of feature variation and leverage the benefits of a graph-based approach, representing spatial and spectral information. Then, we perform a stacking technique for temporal datasets, resulting in multiple temporal graphs. Such methodology resulted in a temporal multispectral GNN–GAN for generating images following back and forecasting results (see Fig. 1). Through integrating GNN and GAN architectures, we offer a novel approach to address the challenges of temporal data scarcity, ultimately facilitating more robust and accurate analysis in data-driven applications. We tested our approach on crop disease forecasting datasets. The summarized main contributions of our work are as follows.

1) Seamless integration of a GAN and GNN architecture for multispectral and temporal data augmentation, leading to an integration that bolsters the accuracy and coherence of prediction models trained with a combination of real and synthetic data.
2) Introduction of a dual-generator strategy coupled with a novel discriminator designed to evaluate predicted multispectral temporal graphs generated by a pretrained GNN model. Such discriminator critically assesses the disparities between predicted graphs at an instant $t \pm 1$ and the corresponding real graphs, enhancing the model's grasp of temporal relationships in synthetic data generation.
3) Experimental validation of our enhanced GNN–GAN framework showcasing superior performance in generating precise multispectral temporal graphs. The model maintains temporal coherence and accuracy metrics, surpassing previous approaches and affirming its efficacy in producing accurate synthetic multispectral temporal data.

The rest of this article is organized as follows. Section II compares the proposed GNN–GAN integration with related work. In Section III, the proposed model is described in detail. Section IV describes the dataset and experimental results. Finally, Sections V and VI discuss the findings and conclude this article, respectively.

II. Related Works

Several studies have explored the fusion, data imputation, and scene reconstruction of multispectral data in remote sensing applications, demonstrating advancements and the importance of temporal resolution in downstream tasks (e.g., prediction and classification). Notably, these investigations encompass diverse approaches ranging from spectral–temporal response surfaces (STRSs) to generative models for image fusion and predictive modeling. However, the use of time-series-based synthetic imagery imputation to improve the accuracy of downstream tasks, such as plant disease detection, remains unaddressed in existing studies. For example, the study by Gevaert et al. [22] focuses on constructing STRS using Bayesian theory to fill in missing spectral data within multispectral imagery. It emphasizes the impact of imputation techniques on vegetation indices such as OSAVI and MCARI, underscoring the importance of temporal data imputation in precision agriculture. Similarly, Azarang and Kehtarnavaz [10] explore a generative model for multispectral image fusion but do not specifically address temporal data imputation strategies. Their focus remains on unsupervised learning and multi-objective loss functions for image fusion.

Using GAN-based approaches, Jiang et al. [11] introduced a heterogeneously integrated framework utilizing deep residual cycle GANs for image fusion across spatial, temporal, and spectral domains. While tackling fusion challenges comprehensively, their work does not explicitly tackle temporal imputation to enhance predictive capabilities. Yang et al. [23] present the YPM-GAN-GAT model, leveraging GNN and GAN models for corn yield prediction. While resembling the current proposal, they diverge by emphasizing trait attribute completion for yield estimation, not considering temporal imputation and imagery generation approaches—elements pivotal in the current paper's scope. In a similar direction, TimeGAN [16] and GT-GAN [17] offer capabilities for synthesizing data based on predefined window sizes. These methodologies cater to time-series analysis, particularly emphasizing extensive window sampling spanning days, weeks, or even months. They also highlight the necessity for additional temporal data, while not extending their scope

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to encompass generating synthetic imagery or addressing scenarios with low temporal resolution. Our method relies on analyzing the spectral–temporal time series but for generating new imagery, which might be used, for example, to train a machine learning model in a new downstream task. Wu et al. [24] designed a similar method, but for reconstructing missing areas for time-series images, using a spectral–temporal patch-based approach through multitemporal image segmentation. This process effectively enhanced accuracy and precision by predicting mean and standard deviations through linear regression and integrated textural information from clear, temporally adjacent patches. In our approach, we generate entirely new images corresponding to specific instants within a given time period.

III. Method

A. Problem Statement

The low temporal resolution of remote sensing data is one of the factors that affect prediction models' accuracy [25], especially in crop analysis [26], and the challenge that the proposed method addresses. In this section, we formalize the problem of low temporal resolution in multispectral imagery.

Consider a set S_t of multispectral imagery captured at different times, denoted by $\{t_1, t_2, t_3, \ldots, t_N\}$, where N represents the total number of timestamps observed. Furthermore, for this multispectral imagery dataset, there is a nonconstant variation in Δ_t between each timestamp. For instance, between t_1 and t_2, Δ_t could be y, while between t_2 and t_3, Δ_t might be $y+y$. This situation is especially common in remote sensing data, where UAV-based data collection often occurs at irregular intervals without a fixed time window [27]. Similarly, satellite data can exhibit nonuniform temporal dimensions due to cloud cover or orbital specifics, affecting the consistency of observations in a given study [28].

In this context, the challenge is to effectively augment the sample size of S_t so it improves the accuracy of downstream tasks (e.g., classification and regression).

B. Solution Overview and Architecture

To address the low-temporal resolution problem, we rely on two foundations: 1) the capability of graph structures to represent the temporal dimension of multispectral imagery and 2) the capability of GANs to generate multispectral synthetic imagery. Our solution considers such temporal dimension and generates synthetic imagery for a specified instant of time. The graph representation serves as input to a temporal predictive GNN model, which aims to learn the temporal relationships between an image instance at the instant t and the subsequent instant at time $t+1$, considering the Δ_t (see the latent correlation in Fig. 2). The passage of time, for example, the number of days between a graph g_a and a graph g_b, is taken into account to define this correlation.

The combination between the GNN model as an equalizer for the GAN model is illustrated in Fig. 2, containing the architecture of our method. In this figure, X_t and $X'_{\pm t}$ represent the real input multispectral temporal images and the synthetic multispectral temporal images generated as output of our pipeline, respectively. The elements of $X'_{\pm t}$ are the synthetic temporal imagery generated for an instant t. Note that our geospatial "time machine" capability is achieved simply by altering the direction of navigation through the available temporal data instants.

In remote sensing, the bands of a multispectral image [e.g., blue, red, green, near-infrared (NIR), and red edge (RE)] and the covariance between them (according to the target study area) play a crucial role. In our method, we imposed a physics-based constraint, specifying that if covariance exists between two bands of X, X' must reproduce the same covariate behavior in its pixel values. The proposed method explicitly encodes such differences in covariate behavior in the input and synthetically generated images as a loss in the discriminator model.

Utilizing the vegetation analysis as an example (see Section IV-A), multispectral images exhibit characteristics of absorbing red light and reflecting NIR light prominently. This is evidenced by absorption valleys, reflection, and scattering plateaus observed at NIR wavelengths. The RE denotes the wavelength range between 680 and 750 nm, where the spectral reflectance of plants undergoes a sharp transition, indicating responses to biotic or abiotic stressors [29]. Thus, the integrated GNN–GAN model compels its underlying latent space to adhere to the established physical constraint governing the relationship between NIR and RE [30].

In this context, for each input/source image x belonging to X, its covariance coefficients c should also manifest in any corresponding x' element belonging to X'. To achieve this, our generative multispectral model includes a generator denoted as $G(z; \theta_h)$, with z being the input noise vector and θ_h signifying the generator's parameters. The objective of G is to generate the elements of X' while adhering to the coefficients of C and undergoing evaluation by discriminators D.

For each generated X', there are three discriminators, namely, $D_1(X; X'; \theta_1)$, $D_2(X_{\text{spectral}}; X'_{\text{spectral}}; \theta_2)$, and $D_3(G(X); G(X'); \phi)$. These discriminators are responsible for assessing the distribution, covariance, and graph distance of X' and X, respectively. Here, X_{spectral} and X'_{spectral} refer to the spectral profile of X and X', and θ_1, θ_2, and ϕ denote their respective parameters.

C. Multispectral Temporal GNN–GAN Model

The model proposed in this article consists of seven main modules: optimizer, spectral regularization, two generators, and three discriminators, as depicted in Fig. 2. To facilitate the comprehension of our model as well as for notation simplicity, consider X and X' arrays $(N, N, 5)$, in the sense that five-band multispectral imagery will be considered, and N is the imagery size.

1) Generators: Our model consists of two generators: $G_{\text{multispectral}}$ and G_{temporal} (see Generators A and B, respectively, in Fig. 2). $G_{\text{multispectral}}$ takes random noise n as input, with latent space values adjusted by the optimizer O and generates synthetic multispectral imagery $x'_{\text{multispectral}}$. It aims to produce samples that deceive the discriminators into classifying $x'_{\text{multispectral}}$ as real. Following the architecture

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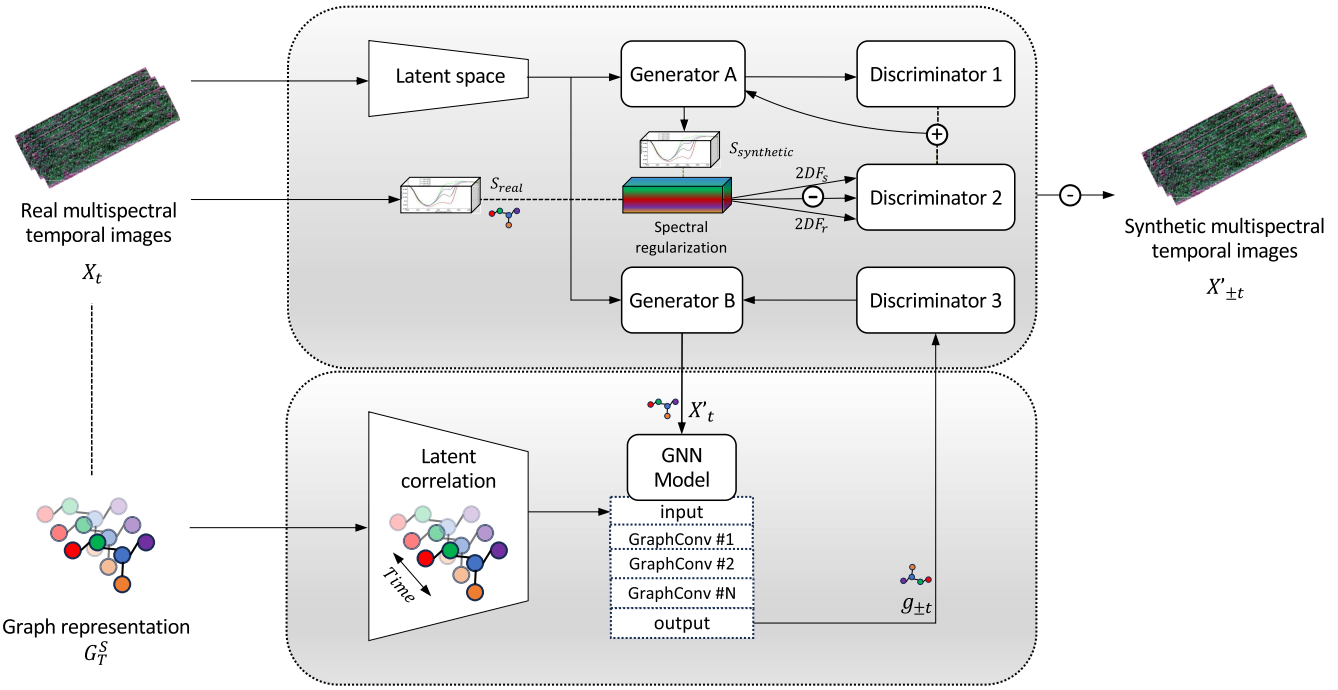


Fig. 2. Illustrated overview with the architecture of the proposed GNN–GAN model showcasing the role of the third discriminator that evaluates the predicted multispectral temporal graph generated by a pretrained GNN model. The GAN's generator components aim to minimize the disparity between the graph representing the real image at time $t \pm 1$, and the synthetic data generated at time $t \pm 1$, emphasizing the learning process required by the generator to achieve this minimization.

of DCGAN [31], $G_{\text{multispectral}}$ employs convolutional layers to transform n into synthetic imagery and utilizes deconvolutional layers to upsample the vector n into a higher dimensional representation.

G_{temporal} specializes in handling temporal data, following a design similar to $G_{\text{multispectral}}$. It takes noise input to generate synthetic sequences x'_{temporal} specifically suited for temporal aspects. The architecture employs customized convolutional and deconvolutional layers tailored for processing temporal data, aiming to create high-fidelity temporal sequences. It is important to note that the output from G_{temporal} is further transformed into a graph format and evaluated by the pretrained GNN model.

2) Discriminators: For each set of synthetic samples X', our GNN–GAN model incorporates three discriminators, each serving distinct roles: 1) assessing the proximity of each $x' \in X'$ to the real data x; 2) evaluating the spectral divergence between x and x'; and 3) considering the temporal graph predictions generated by a pretrained GNN model for x' at time $t \pm 1$. The discriminator $D_1(x; x'; \theta_s)$ aims to determine the likelihood that x' originates from the real data distribution rather than being synthetically generated. Simultaneously, $D_2(x; x'; \theta)$ receives the spectral information of x and x', assessing the dynamic spectral distance to identify if x' belongs to the spectral profile of x. Meanwhile, the discriminator $D_3(x'; G(x); \phi)$ critically evaluates the temporal graph prediction $[G(x)]$ of the synthetic data x' generated by the GAN's generator, emphasizing the discrepancy between the predicted multispectral temporal graph at time $t \pm 1$ and the real graph at the same time frame.

Thus, the objective of our model with two generators and three discriminators is to find equilibrium by solving the following minimax problem:

$$\begin{aligned}\min_h \max_s V(D_1, D_2, D_3, G_1, G_2) \\ = \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D_1(x) + \log D_2(x) + \log D_3(x)] \\ + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D_1(G_1(z))) + \log(1 - D_2(G_1(z))) \\ + \log(1 - D_3(G_2(z)))]\end{aligned} \tag{1}$$

where

1) $\mathbb{E}_{x\sim p_{\text{data}}(x)}$ expectation over real data samples;
2) $\mathbb{E}_{z\sim p_z(z)}$ expectation over the gradients for the spectral vectors sampled from the prior distribution $p_z(z)$;
3) $D_1(x)$, $D_2(x)$, and $D_3(x)$ output probabilities of the three discriminators for real data x;
4) $D_1(G(z))$, $D_2(G_1(z))$, and $D_3(G_2(z))$ output probabilities of the three discriminators for synthetic data generated by the generators G;
5) $V(D_1, D_2, D_3, G_1, G_2)$ value function of the minimax game, maximizing with respect to the discriminator parameters $(\theta_1, \theta_2, \theta_3)$ and minimizing with respect to the generators parameters (θ_h).

In the proposed GNN–GAN model, the third discriminator (D_3) processes the generated fake data through a pretrained GNN model, producing a graph representation structured as an N-tuple, with N being the number of multispectral bands and segmented into S segments representing nodes that group significant pixels. Notably, the resultant graph from the synthetic imagery is based on the output of the GNN model trained to return the next or previous probable graph

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according to a time t. Subsequently, the graph derived from the synthetic imagery is compared with the graph obtained from real data by calculating a distance L. This distance measurement, representing dissimilarity or similarity between graphs, utilizes the Euclidean distance computation based on node attributes or positions within the graph structure [32]. The minimization problem for D_3 to minimize the Euclidean distance between corresponding node attributes or positions in the G_{real} and $G_{\text{synthetic}}$ graphs is represented as follows:

$$L = \min_{G_{\text{real}}, G_{\text{synthetic}}} \sum_{i=1}^{N} \left\| G_{\text{real}}(i) - G_{\text{synthetic}}(i) \right\|^2. \tag{2}$$

Here, N represents the total number of nodes in the graph and $G_{\text{real}}(i)$ and $G_{\text{synthetic}}(i)$ denote the attributes or positions of node i in the real and synthetic graphs, respectively. This minimization problem aims to reduce the dissimilarity between the structures represented by these graphs.

3) Graph-Based Representation and GNN Model: Utilizing a GNN for back and forecasting involves leveraging the graph representation of imagery to perform predictive tasks, capturing complex relationships within the image data.

Representing imagery as graphs, instead of spectral representations, for back and forecasting is justified by the inherent advantages of graph-based approaches in preserving spatial relationships, accommodating irregular data structures, and leveraging graph-based neural networks for predictive tasks. Graph representations maintain crucial spatial context by capturing pixel-to-pixel connections as nodes and edges, allowing for more accurate analysis of intricate relationships within the image—such pixel-to-pixel association is missed in spectral profiles. Thus, leveraging graph-based neural networks like GCNs facilitates forecasting by exploiting graph topology and capturing nonlocal relationships, enabling more nuanced and context-aware predictions. This choice empowered our predictive model to handle spatial dependencies and irregularities within the image data, potentially leading to more robust and accurate forecasting results.

The proposed methodology applies the simple linear iterative clustering (SLIC) algorithm [33] for superpixel to create the graph representation from multispectral imagery segmentation. This step creates a more meaningful and easier representation to analyze [34] compared with the original set of pixels. Superpixels are compact, contiguous regions in an image that combine pixels with similar properties, such as color or texture. The objective is to segment an input image into a specified number of superpixels while additionally connecting neighboring superpixels in the graph by adding edges between them.

Let I be the input dataset, where I has dimensions (length, height/width, channels/features). The SLIC algorithm aims to partition the dataset into K coherent segments or superpixels. It can be described as shown in Algorithm 1.

The iterative process of assignment and update continues until convergence, minimizing the compactness and adherence to segment boundaries while considering the desired number of segments K, i.e., time instants. Fig. 3 illustrates the outcome of the transformation achieved using our method for a plant disease detection problem. The real data comprises five instants, with each instant being a set of multispectral data collected after n days of plant growth.

Algorithm 1

Require: Dataset I, Number of clusters K
Ensure: Cluster centers C_k
Initialization: ▷ Initialize K cluster centers or seeds $C_k = (l_k, a_k, b_k, \dots)$
Initialize K cluster centers or seeds $C_k = (l_k, a_k, b_k, \dots)$ ▷ where k represents each cluster center and $l_k, a_k, b_k, \dots$ denote feature values (e.g., in the case of images, l, a, b might represent color channels)
Assignment Step:
for each pixel p in the dataset I **do**
Compute the distance between p and each cluster center C_k using a spatial and feature space distance metric
Assign p to the nearest cluster center based on these distances
end for ▷ For each pixel p in the dataset I, compute the distance between p and each cluster center C_k using a spatial and feature space distance metric. Assign p to the nearest cluster center based on these distances.
Update Step:
Recalculate the cluster centers C_k by taking the mean of all pixels assigned to each cluster
Ensure adherence to compactness and boundary constraints

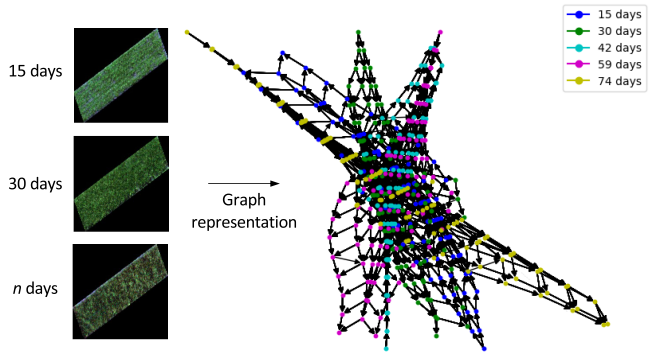


Fig. 3. Representation instance of our multispectral temporal dataset as a set of graphs.

The formulaic representation of SLIC highlights the initialization of cluster centers, the assignment of pixels to clusters based on proximity, and the iterative refinement of cluster centers to achieve coherent segments within the dataset.

The discriminator D_3 being a model based on a GNN that operates on graph-structured data, which is specifically designed to process both *spatial* and *temporal* features. Our GNN architecture is composed of multiple graph convolutional layers and is structured as follows.

The *input layer* consists of a graph $G = (V, E)$, where V represents the set of nodes, and each node v_i has a feature vector $x_i \in \mathbb{R}^{d_{\text{in}}}$; where d_{in} is the input feature dimension and E represents the set of edges connecting the nodes in the graph. In addition, each node may have a temporal feature, such as a *timestamp* or *day*, denoted by $t_i \in \mathbb{R}$, which is

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incorporated into the node features to account for temporal dynamics. The *graph convolutional layers* is the core of our architecture and consists of L stacked *graph convolutional layers*. Each layer l is a graph convolutional operation defined as follows:

$$h_i^{(l)} = \mathrm{GCN}\left(\mathbf{A} \cdot h_i^{(l-1)}\right)$$

where $h_i^{(l)}$ represents the feature vector for node i at layer l. $\mathbf{A}$ is the adjacency matrix of the graph, which encodes the connectivity between nodes. The convolutional operation aggregates information from neighboring nodes and applies an activation function (typically ReLU) after each convolution to introduce nonlinearity.

The first graph convolutional layer takes the input feature dimension d_{in}, and subsequent layers operate with a hidden feature dimension d_{hidden}. Each layer transforms the node features h_i to a higher level representation, and the number of layers L allows the model to capture complex graph dependencies over multiple hops.

If temporal features t_i are available (e.g., representing the *day* or *timestamp* for each node), these features are concatenated with the spatial features before the graph convolutions. This step allows the model to jointly learn both the spatial and temporal dependencies. The concatenation is performed as follows:

$$x_{\text{new}} = [x_i, t_i]$$

where $x_i \in \mathbb{R}^{d_{\text{in}}}$ is the original spatial feature and $t_i \in \mathbb{R}$ is the temporal feature, resulting in a combined feature vector $x_{\text{new}} \in \mathbb{R}^{d_{\text{in}}+1}$.

The final output is computed using another *graph convolutional layer*, which maps the hidden node representations to an output space of dimension d_{out}. This layer is defined as follows:

$$y_i = \mathrm{GCN}\left(\mathbf{A} \cdot h_i^{(L)}\right)$$

where $\mathbf{A}$ is the adjacency matrix and $y_i \in \mathbb{R}^{d_{\text{out}}}$ represents the final output for node i. This output can correspond to different tasks, such as: 1) node classification, where the output is a classification label for each node and 2) node regression, where the output is a continuous value for each node.

IV. Evaluation

The proposed method was evaluated from three different perspectives, described in Sections IV-C–IV-D. First, we analyzed the performance of our method compared with other approaches and assessed the similarity between the synthetic temporal imagery and ground-truth data. Then, we explored if synthetic-based temporal vegetation indices would follow a real-world behavior. Last, we observed the impact of using our GNN–GAN integration in a scenario with limited temporal data and the performance of a downstream classification task when using the generated synthetic temporal data for training. The GTM was implemented using Python, PyTorch, and TensorFlow. Interested readers can access the code at: https://github.com/RSL-SLU/gtm. The next subsection describes the dataset utilized in our experiments.

A. Dataset

For training and evaluating the proposed GNN–GAN architecture, we utilized a multispectral imagery dataset of spring wheat collected on multiple dates at Chacabuco (−34.64; −60.46), Argentina. The data collection experiment consisted of growing three varieties of spring wheat, categorized as an intermediate-late-long cycle, intermediate cycle, and short cycle, in a total of 700 field plots. The plots were arranged in a grid pattern with 35 rows and 20 columns within the field. Each plot had dimensions of 1.2 × 5 m and consisted of seven rows of wheat plants. Spring wheat was planted on June 19, 2021, and harvested on December 16, 2021. The soil type at the site was classified as fine-silty, mixed, and thermic Typic Argiudoll according to USDA-Soil Taxonomy V. 2006. Throughout the growing season, uniform field management practices were implemented for all field plots. Insecticide, herbicide, and fertilizer were applied as needed to ensure consistent plant health and growth. However, no fungicide was used in the experiment, as the focus was on studying the impact of the biotic stress caused by the occurrence of yellow rust disease.

UAV flights were conducted using the DJI Phantom 4 Pro [2], [3] as the primary aerial platform for collecting remotely sensed data from the standing crop. Equipped with a multispectral sensor capable of capturing five discrete spectral bands blue (450 ± 16 nm), green (560 ± 16 nm), red (650 ± 16 nm), RE (730 ± 16 nm), and NIR (840 ± 26 nm), the UAV was prepared for high-resolution imaging. A sunlight irradiance watcher was also connected to the sensor to ensure accurate measurements. The UAV was programmed to fly at a low altitude of 20 m from the surface to achieve a high spatial resolution, resulting in a pixel resolution of 1.04 cm.

The aerial data collection was designed with a temporal dimension to monitor the crop at different stages of growth. The UAV conducted five flights throughout the growing stages, commencing with the first flight on August 30, 2021, during the tillering stage, and concluding with the last flight on November 17, 2021, during the flowering stage.

B. Similarity and Performance Analysis

To assess the similarity between the synthetic temporal imagery and ground-truth data, we trained the GNN model using samples from *three of five* different timestamps of our dataset, allowing $D_3(x'; G(x); \phi)$ to learn the latent correlations between these points. Then, we parameterized the time (i.e., the number of days) for the training input of the GAN model, treating it as a feature received by $G(x)$. To prevent data leakage to D_3 (our temporal discriminator), two instants (30 and 42 days) were excluded from the training dataset. This exclusion ensured that we could evaluate the effectiveness of our synthetic temporal image generation method for specific time instants, such as 30 days, without including these instants in the training set. In this scenario, Fig. 4 illustrates the R^2 values of mean spectral profiles derived from synthetic temporal imagery generated for an instant t from $t+1$, where t is limited to instances with comparable ground-truth data. The findings indicate that the model tends to predict higher values for

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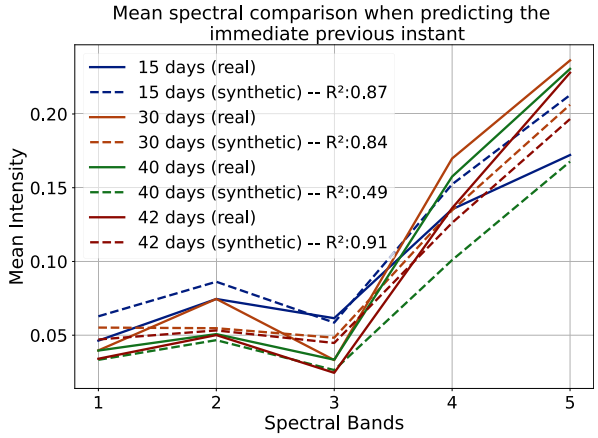


Fig. 4. Ground-truth dataset consists of multispectral imagery collected on five different dates. This chart depicts the prediction for the instant (t) $t-1$.

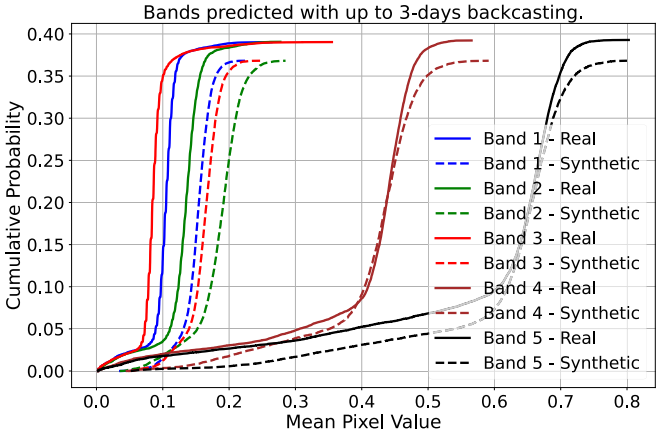


Fig. 5. Probability analysis of each band when backcasting the imagery for three days in the past using the proposed method. The model achieves a cumulative error of ∼20% when predicting the pixel values for bands 1–3.

bands 4 and 5 optimistically. In addition, it exhibits suboptimal performance in capturing significant variations occurring over a short time span, such as the variations observed in spectral profiles between 40 and 42 days.

Expanding the time window to $t-3$ and computing the mean pixel values for all available ground-truth data, Fig. 5 present the distribution of pixel values per band and the probability of GMT accurately predicting each band's value. As anticipated, the model demonstrates superior performance in predicting pixel values for bands that exhibit minimal temporal variation.

In addition to verifying spectral profile similarity, we performed a downstream pixel-based classification task to assess model performance. Specifically, we used SHapley Additive exPlanations (SHAPs) to interpret the predictions by attributing feature importance values from two convolutional neural network (CNN) models: one trained exclusively on real samples and another trained on a combination of real and synthetic temporal data. The primary goal was to identify diseased pixels within an unhealthy plot, with SHAP values helping to highlight which pixels were most strongly associated with the diseased class. We chose the CNN model because it is widely used for classification tasks in the remote sensing field [35]. By comparing the SHAP values across both models,

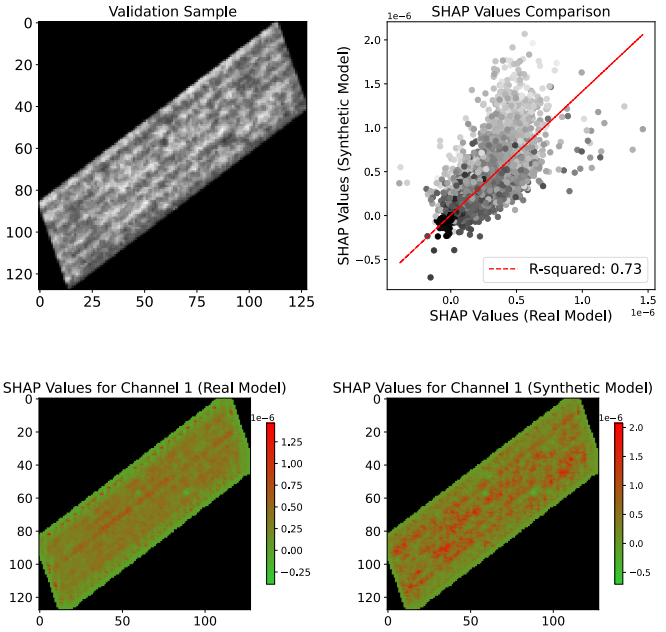


Fig. 6. At the top, the scatter plot compares SHAP values from the real and temporal synthetic-based CNN models for the same validation sample. Each point represents a pixel, with color indicating pixel intensity. The red dashed line represents the linear regression fit with an R^2—0.73, indicating a reasonable correlation between the SHAP values from both models. The two images at the bottom also indicate that the models identified the same pixel regions as disease-infected areas; however, the result for the model trained with temporal synthetic data (bottom right) is spatially clearer.

we evaluated the consistency of pixel importance and assessed the extent to which synthetic data could replicate the behavior of the real-data-driven model. Fig. 6 illustrates the outcome of this experiment, wherein the CNN models accurately classify an unhealthy validation sample. The model trained with synthetic temporal samples offers a clearer spatial delineation of potential diseased plants, indicated by red pixels in the chart, compared with the more blurred region from the model trained solely on real imagery.

Note that we utilized expert-labeled data as ground truth to determine whether a plot was diseased. Each point in the scatter plot at the top of Fig. 6 represents a pixel, with the color indicating pixel intensity. The red dashed line, representing a linear regression fit with an R^2 of 0.73, suggests a strong correlation between the SHAP values from both models. This indicates that the CNN model trained on synthetic imagery identified the same pixel regions relevant to plot health as the model trained on real imagery.

The images at the bottom of the figure demonstrate that both models—one trained exclusively on real data and the other trained on synthetic temporal data—identify the same pixel regions as disease infected. Notably, the model trained on synthetic temporal data (bottom right) offers a sharper and more distinct spatial delineation of these areas.

Finally, we evaluated the efficacy of our proposed model in comparison to similar models documented in the literature, namely, TimeGAN, GT-GAN, and an LSTM-based GAN implementation (Fig. 7). We conducted an assessment of synthetic samples generated by these models, analyzing

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TABLE I

COMPARISON OF DIFFERENT MODELS BEING USED AS SPECTRAL–TEMPORAL PREDICTORS WHEN GENERATING TEMPORAL SYNTHETIC SAMPLES

Models	FID ↓	BC ↓	IoU ↑
Baseline	0.21	0.30	0.04
GTM (*ours*)	**0.19**	**0.54**	0.20
TimeGAN [16]	0.22	0.63	0.19
GT-GAN [17]	0.31	0.68	**0.24**
LSTM	0.91	0.73	0.03
PlantPlotGAN [36]	0.07	0.51	0.12

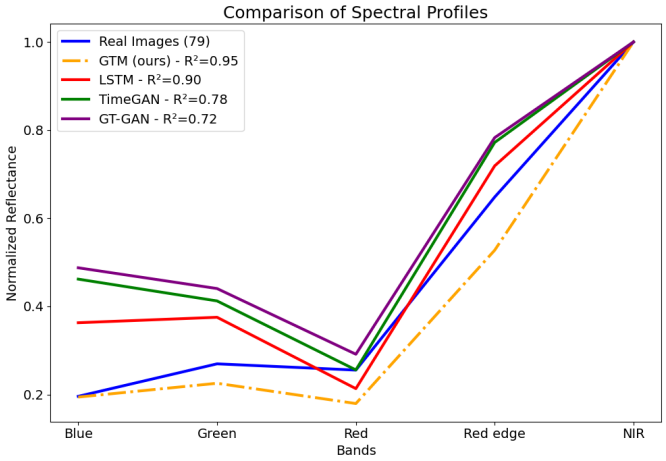


Fig. 7. Comparison of the mean spectral profile generated by GTM with those produced by state-of-the-art deep learning models (e.g., LSTM and TimeGAN) for predicting the final stage in our plant growth dataset.

three widely recognized metrics: Fréchet inception distance (FID), Bhattacharyya coefficient (BC), and intersection over union (IoU). All models were trained for 50 epochs, with ten steps per epoch, using a learning rate of 0.0002 and the Adam optimizer. In addition, while the metrics for PlantPlotGAN only consider predictions made for the same instant t, the other models attempt to predict plot appearances with a backcasting window of seven days. The results of this evaluation are summarized in Table I. GTM demonstrates superior performance compared with other time series-based models in two key metrics: FID and BC. However, it does not achieve the highest Intersection result from the analysis. We attribute GTM's superior performance in FID and BC to the high-resolution temporal data dependency present in the other models, as well as the physics-constrained spectral value generation implemented in GTM (as discussed in Section III). On the other hand, the pixel cumulative probability for bands 4 and 5 obtained from the samples generated by GTM negatively impacted its IoU result. It is worth noting that the baseline for these experiments consists of rotated samples, all considering the same instant t. This helps isolate and fairly analyze the impact of the synthetic samples on forecasting.

C. Feature Analysis Between Temporal Synthetic and Real Imagery

We evaluated whether the temporal imagery generated from GMT could exhibit behavior similar to real imagery when analyzed through the perspective of vegetation indices. We performed feature engineering to extract 43 vegetation indices (VIs) selected from the literature [12] (see supplementary material). The VIs were extracted by calculating the mean value over a region for each of the 700 plots in our dataset (see Section IV-A). From these 43 VIs, we arbitrarily selected five indices and analyzed them using a window size of up to seven days during backcasting. We then compared the results with our ground-truth data as follows.

1) *Normalized Difference Vegetation Index (NDVI):* NDVI measures the amount of vegetation present in an area. It is calculated from the difference between NIR and red light reflectance. Higher NDVI values generally indicate healthier and more abundant vegetation.
2) *Normalized Difference Water Index (NDWI):* NDWI is used to detect the presence of water in vegetation. It is calculated from the difference between NIR and green light reflectance. Higher NDWI values typically indicate the presence of water.
3) *Green NDVI (GNDVI):* GNDVI is similar to NDVI but uses green light instead of red light. It is also a measure of vegetation density and health.
4) *Meris Terrestrial Chlorophyll Index (MTCI):* MTCI is a vegetation index sensitive to chlorophyll content in leaves and helps in assessing plant health and stress.
5) *Red Edge NDVI (RE-NDVI):* RE-NDVI is a modified version of NDVI that uses the red-edge bands of satellite imagery. It is particularly useful for monitoring vegetation stress and changes in plant phenology.

These indices offer a comprehensive view of the physiological state and health status of the plants.

Fig. 8 shows the result of this analysis. The temporal synthetic plots were able to achieve high similarity with the real ones, which suggests that—even for dates without ground-truth data—our model is capable of generating samples that reproduce the appearance (from the spectral and spatial perspective) of real imagery.

TABLE II

COMPARISON OF DIFFERENT COMPONENTS ENABLED IN OUR METHOD

Models	FID ↓	BC ↓	IoU ↑
Baseline	0.21	0.30	0.04
GTM	**0.19**	**0.54**	**0.20**
GTM $-G_{spectral}$ $+G_{temporal}$	0.41	0.70	0.17
GTM $+G_{spectral}$ $-G_{temporal}$	0.34	0.68	0.18
GTM (no GNN component)	0.08	0.49	0.12

D. Impact of Synthetic Temporal Data on Prediction Models

We also designed an experiment to analyze whether synthetic temporal data could improve the accuracy of prediction models when analyzing VIs (see supplementary material). In this experiment, we generated synthetic temporal imagery for each date of the ground-truth data (mostly for the minority class—healthy plots), always considering the immediate previous instant, and extracted the 43 VIs. Afterward, we verified if the classification of plant disease (annotated at the last date from the ground truth) would improve using early-stage data (real and synthetic). We also compared the results with

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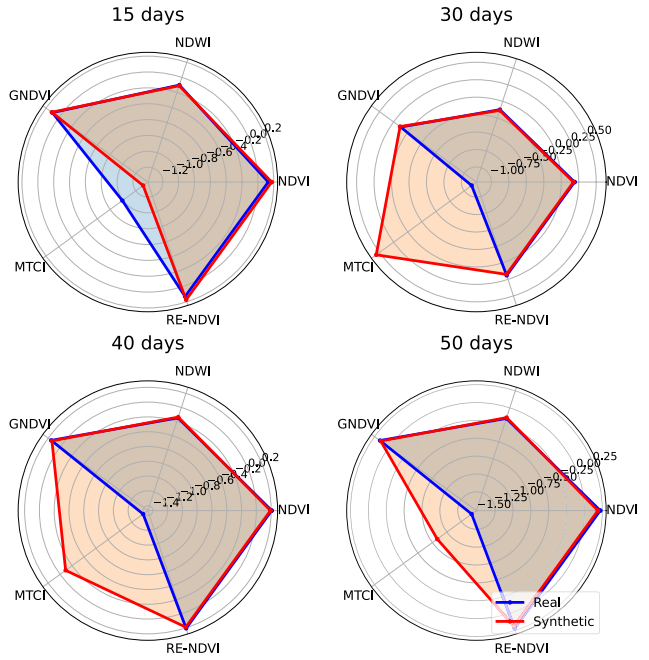


Fig. 8. Comparison of vegetation indices for each plant growth stage derived from real and synthetic data with a backcasting window of up to seven days.

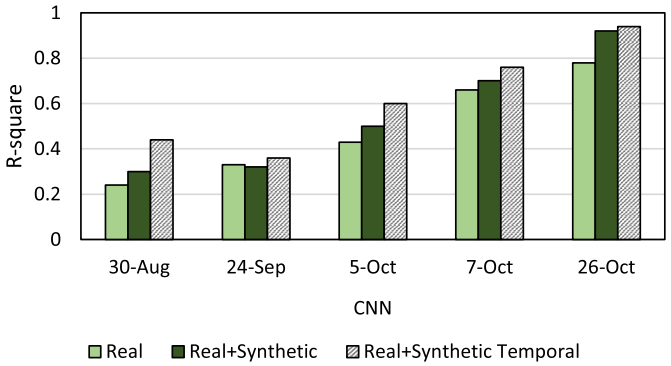


Fig. 9. Analyzing the accuracy of a CNN-based classification forecast model using different datasets for training the predictor, including the training set that combines the synthetic temporal data generated by the proposed method. The validation set used was the same in all scenarios. As the chart depicts, the utilization of synthetic temporal data helped improve accuracy in the early stages of plant growth.

those obtained in a previous work [36], which focused solely on balancing the dataset (i.e., healthy and unhealthy) using synthetic data. The validation set was the same in all scenarios.

The results of this experiment indicate that our GNN–GAN model is capable of generating realistic temporal synthetic data and improving the accuracy [see (3)] of a regular CNN by up to 53% (see Fig. 9) at the earliest stages of plant growth when predicting plant disease. Such finding suggests a significant benefit of performing scene reconstruction or temporal data augmentation using GAN- and GNN-based generative techniques. The noise in the results from September 24 is due to weather problems that occurred during UAV data collection (which jeopardized the imagery quality and image calibration process). This explains why the downstream task did not improve its forecast accuracy at this plant growth stage—a behavior not observed when the model was trained with samples from other stages

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100\%. \tag{3}$$

E. Ablation Study

To better understand the contributions of the individual components of our GTM, we performed an ablation study. This study isolates or modifies specific aspects of the GTM framework to evaluate their impact on the model's performance. Table II presents the results of various model configurations, each highlighting a different component of GTM. We used the same evaluation metrics (i.e., FID, BC, and IoU) that were previously applied to compare GTM with state-of-the-art methods. The task in this study was consistent: generating synthetic images for an instant of seven days in the past.

As anticipated, disabling certain components led to a decline in performance. However, the ablation study results emphasize the critical role of both generators in achieving optimal performance. Notably, the G_{spectral} component demonstrated a slight advantage, suggesting its importance in improving the quality of synthetic data.

V. Discussion

The evaluation of the proposed method encompasses various aspects, each shedding light on different facets of its performance and potential applications. Here, we discuss the findings and implications of each evaluation perspective.

The comparison of our method with existing approaches forms the cornerstone of our evaluation. By assessing the performance against other methods and examining the similarity between synthetic temporal imagery and ground-truth data, we gauged the efficacy and realism of our approach. Notably, excluding certain time instants during training to prevent data leakage underscores our method's robustness in capturing temporal dynamics without relying on specific data points.

The statistical analysis, including FID, Battacharayya, and intersection metrics (as presented in Table II), provides a quantitative performance assessment relative to other methods. These metrics provide a detailed view into the fidelity of the generated imagery compared to ground truth, enabling a comprehensive evaluation of our approach's efficacy.

Feature analysis, particularly concerning vegetation indices, is crucial for understanding the behavior of synthetic imagery relative to real-world data. The comprehensive exploration of 43 VIs, coupled with the comparison of selected indices using a range of window sizes during backcasting, could demonstrate the capability of our method to replicate real-world vegetation dynamics.

The comparison of indices such as NDVI, NDWI, GNDVI, MTCI, and RE-NDVI, along with the radar chart visualization presented in Fig. 8, illustrates the high degree of similarity between synthetic and real imagery. This suggests that our model effectively captures vegetation dynamics over time,

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TABLE III
VEGETATION INDEX FORMULAS AND REFERENCES

No.	Index Names	Index Formula
1	Normalized difference vegetation index (NDVI)	$\mathrm{NDVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}}$ [37]
2	Green normalized difference vegetation index (GNDVI)	$\mathrm{GNDVI} = \frac{\mathrm{NIR}-\mathrm{G}}{\mathrm{NIR}+\mathrm{G}}$ [38]
3	Ratio vegetation index (RVI)	$\mathrm{RVI} = \frac{\mathrm{NIR}}{\mathrm{R}}$ [39]
4	Green chlorophyll index (GCI)	$\mathrm{GCI} = \frac{\mathrm{NIR}}{\mathrm{G}} - 1.0$ [40]
5	Red-green ratio vegetation index (RGVI)	$\mathrm{RGVI} = \frac{\mathrm{R}}{\mathrm{G}}$ [41]
6	Difference vegetation index (DVI)	$\mathrm{DVI} = \mathrm{NIR} - \mathrm{R}$ [42]
7	Soil-adjusted vegetation index (SAVI)	$\mathrm{SAVI} = \left(\frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}+0.5}\right) \times (1.0 + 0.5)$ [43]
8	Modified soil-adjusted vegetation index (MSAVI)	$\mathrm{MSAVI} = 0.5 \times \left(2.0 \times \mathrm{NIR} + 1.0 - \sqrt{(2.0 \times \mathrm{NIR} + 1.0)^2 - 8.0 \times (\mathrm{NIR} - \mathrm{R})}\right)$ [44]
9	Optimized soil-adjusted vegetation index (OSAVI)	$\mathrm{OSAVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}+0.16}$ [45]
10	Renormalized difference vegetation index (RDVI)	$\mathrm{RDVI} = \sqrt{\frac{(\mathrm{NIR}-\mathrm{R})^2}{\mathrm{NIR}+\mathrm{R}}}$ [46]
11	Triangular vegetation index (TVI)	$\mathrm{TVI} = 60.0 \times (\mathrm{NIR} - \mathrm{G}) - 100.0 \times (\mathrm{R} - \mathrm{G})$ [47]
12	Transformed soil-adjusted vegetation index (TSAVI)	$\mathrm{TSAVI} = \frac{a \times (\mathrm{NIR} - a \times \mathrm{R} - b)}{a \times \mathrm{NIR} + \mathrm{R} - a \times b}$ [48]
13	Perpendicular vegetation index (PVI)	$\mathrm{PVI} = \frac{\mathrm{NIR} - a \times \mathrm{R} - b}{\sqrt{1 + a^2}}$ [49]
14	Soil-adjusted vegetation index 2 (SAVI_2)	$\mathrm{SAVI}_2 = \frac{\mathrm{NIR}}{\mathrm{R} - \frac{b}{a}}$ [50]
15	Adjusted transformed soil-adjusted vegetation index (ATSAVI)	$\mathrm{ATSAVI} = \frac{a \times (-a \times \mathrm{R} - b)}{a \times \mathrm{NIR} + \mathrm{R} - a \times b + X \times (1 + a^2)}$ [51]
16	Normalized difference water index (NDWI)	$\mathrm{NDWI} = \frac{\mathrm{G}-\mathrm{NIR}}{\mathrm{G}+\mathrm{NIR}}$ [52]
17	Normalized total pigment to chlorophyll a ratio index (NPCI)	$\mathrm{NPCI} = \frac{\mathrm{R}-\mathrm{B}}{\mathrm{R}+\mathrm{B}}$ [53]
18	Simple ratio pigment index (SRPI)	$\mathrm{SRPI} = \frac{\mathrm{B}}{\mathrm{R}}$ [53]
19	Ratio vegetation index (RVI_2)	$\mathrm{RVI}_2 = \frac{\mathrm{NIR}}{\mathrm{G}}$ [54]
20	Modified chlorophyll absorption ratio index (MCARI)	$\mathrm{MCARI} = (\mathrm{RE} - \mathrm{R} - 0.2 \times (\mathrm{RE} - \mathrm{G})) \times (\mathrm{RE}/\mathrm{R})$ [55]
21	Modified chlorophyll absorption ratio index 1 (MCARI_1)	$\mathrm{MCARI}_1 = 1.2 \times (2.5 \times (\mathrm{NIR} - \mathrm{R}) - 1.3 \times (\mathrm{NIR} - \mathrm{G}))$ [56]
22	Modified chlorophyll absorption ratio index 2 (MCARI_2)	$\mathrm{MCARI}_2 = 1.5 \times (2.5 \times (\mathrm{NIR} - \mathrm{R}) - 1.3 \times (\mathrm{NIR} - \mathrm{G})) \times ((2.0 \times \mathrm{NIR} + 1)^2 - (6.0 \times \mathrm{NIR} - 5.0 \times \mathrm{R}) - 0.5)$ [56]
23	Modified triangular vegetation index 1 (MTVI_1)	$\mathrm{MTVI}_1 = 1.2 \times (1.2 \times (\mathrm{NIR} - \mathrm{G}) - 2.5 \times (\mathrm{R} - \mathrm{G}))$ [56]
24	Modified triangular vegetation index 2 (MTVI_2)	$\mathrm{MTVI}_2 = 1.5 \times (1.2 \times (\mathrm{NIR} - \mathrm{G}) - 2.5 \times (\mathrm{R} - \mathrm{G})) \times ((2 \times \mathrm{NIR} + 1)^2 - (6.0 \times \mathrm{NIR} - 5.0 \times \mathrm{R}) - 0.5)$ [56]
25	Ratio of Modified chlorophyll absorption ratio index and Modified triangular vegetation index 2 ($\mathrm{RMCARI}_{\mathrm{MTVI}_2}$)	$\mathrm{RMCARI}_{\mathrm{MTVI}_2} = \frac{(\mathrm{RE}-\mathrm{R}-0.2\times(\mathrm{RE}-\mathrm{G}))\times(\mathrm{RE}/\mathrm{R})}{1.5\times(1.2\times(\mathrm{NIR}-\mathrm{G})-2.5\times(\mathrm{R}-\mathrm{G}))\times((2\times\mathrm{NIR}+1)^2-(6.0\times\mathrm{NIR}-5.0\times\mathrm{R})-0.5)}$ [57]
26	Enhanced vegetation index (EVI)	$\mathrm{EVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+6.0\times\mathrm{R}-7.5\times\mathrm{B}+1.0}$ [58]
27	Datt's chlorophyll content index (DATT)	$\mathrm{DATT} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{NIR}-\mathrm{R}}$ [59]
28	Normalized difference cloud index (NDCI)	$\mathrm{NDCI} = \frac{\mathrm{RE}-\mathrm{G}}{\mathrm{RE}+\mathrm{G}}$ [60]
29	Plant senescence reflectance index (PSRI)	$\mathrm{PSRI} = \frac{\mathrm{R}-\mathrm{G}}{\mathrm{RE}}$ [61]
30	Structure insensitive pigment index (SIPI)	$\mathrm{SIPI} = \frac{\mathrm{NIR}-\mathrm{B}}{\mathrm{NIR}+\mathrm{R}}$ [62]
31	Spectral polygon vegetation index (SPVI)	$\mathrm{SPVI} = 0.4 \times 3.7 \times (\mathrm{NIR} - \mathrm{R}) - 1.2 \times \lvert\mathrm{G} - \mathrm{R}\rvert$ [63]
32	Transformed chlorophyll absorption in reflectance index (TCARI)	$\mathrm{TCARI} = 3.0 \times ((\mathrm{RE} - \mathrm{R}) - 0.2 \times (\mathrm{RE} - \mathrm{G}) \times (\mathrm{RE}/\mathrm{R}))$ [64]
33	Ratio of TCARI and OSAVI ($\mathrm{R}_{\mathrm{TCARI}_{\mathrm{OSAVI}}}$)	$\mathrm{R}_{\mathrm{TCARI}_{\mathrm{OSAVI}}} = \frac{3.0\times((\mathrm{RE}-\mathrm{R})-0.2\times(\mathrm{RE}-\mathrm{G})\times(\mathrm{RE}/\mathrm{R}))}{(\mathrm{NIR}-\mathrm{R})/(\mathrm{NIR}+\mathrm{R}+0.16)}$ [64]
34	Red edge relative index (RERI)	$\mathrm{RERI} = \frac{\mathrm{RE}-\mathrm{R}}{\mathrm{NIR}}$ [65]
35	Normalized difference red edge index (NDRE)	$\mathrm{NDRE} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{NIR}+\mathrm{RE}}$ [66]
36	MERIS terrestrial chlorophyll index (MTCI)	$\mathrm{MTCI} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{RE}-\mathrm{R}}$ [67]
37	Enhanced vegetation index 2 (EVI_2)	$\mathrm{EVI}_2 = 2.5 \times \left(\frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+2.4\times\mathrm{R}+1.0}\right)$ [68]
38	Red edge chlorophyll index (RECI)	$\mathrm{RECI} = \frac{\mathrm{NIR}}{\mathrm{RE}} - 1$ [40]
39	Normalized excess green index (NEXG)	$\mathrm{NEXG} = \frac{2\times\mathrm{G}-\mathrm{R}-\mathrm{B}}{\mathrm{G}+\mathrm{R}+\mathrm{B}}$ [69]
40	Normalized green-red difference index (NGRDI)	$\mathrm{NGRDI} = \frac{\mathrm{G}-\mathrm{R}}{\mathrm{G}+\mathrm{R}}$ [42]
41	Enhanced normalized difference vegetation index (ENDVI)	$\mathrm{ENDVI} = \frac{\mathrm{NIR}+\mathrm{G}-2.0\times\mathrm{B}}{\mathrm{NIR}+\mathrm{G}+2.0\times\mathrm{B}}$ [70]
42	Anthocyanin reflectance index 2 (ARI_2)	$\mathrm{ARI}_2 = \mathrm{NIR} \times \left(\frac{1.0}{\mathrm{G}} - \frac{1.0}{\mathrm{RE}}\right)$ [71]
43	Carotenoid reflectance index 2 (CRI_2)	$\mathrm{CRI}_2 = \frac{1.0}{\mathrm{G}} - \frac{1.0}{\mathrm{RE}}$ [72]

which is essential for various applications such as agriculture monitoring and environmental assessment. The low accuracy in predicting MTCI is explainable by the bands used to calculate it. MTCI's bands have the highest variance when analyzing the mean pixel value (see Fig. 10). Future work could focus on verifying the optimum model size to train the GAN components in order to capture such variance. Replacing the GAN-based generators with diffusion models is also a potential avenue for further investigation.

The experiment assessing the impact of synthetic temporal data on prediction models offers compelling evidence of the utility and effectiveness of our approach. The significant improvement in classification accuracy, especially in the early stages of plant growth, underscores the potential of synthetic temporal data in enhancing the performance of prediction models. This finding has far-reaching implications for applications, such as disease detection and yield forecasting, where accurate and timely data play a crucial role.

Another open issue is splitting the multispectral data into different generator streams in the GAN architecture, hypothesizing that each generator could perform better if associated with a specific multispectral band. Such a hypothesis is out of the scope of the current work.

A. Limitations

We tackled the challenges of temporal data augmentation by leveraging the latent space employed by GANs, primarily focusing on improving the accuracy of subsequent classification tasks using synthetic temporal data. To reduce

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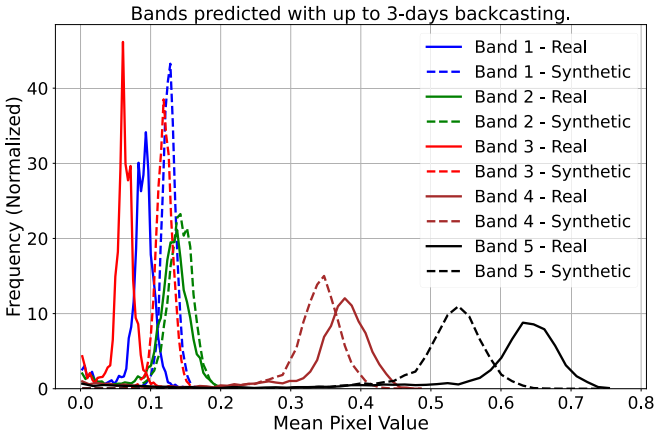


Fig. 10. Analysis of pixel value frequency when performing a backcasting for three days in the past using the proposed method.

computational demands, our input layer comprises randomly generated vectors of size $N100$. Future research could explore the feasibility of inputting specific multispectral plot data and reconstructing its historical context to generate synthetic imagery for a desired past time frame, bearing in mind the high computational resources required for such an approach.

Another aspect requiring attention is the evaluation of additional real-world temporal datasets (e.g., flooding and wildfire). This would help demonstrate the feasibility and generalization of the proposed method across diverse scenarios.

VI. Conclusion

The study introduced the geospatial time-machine concept based on GNNs and GANs to generate temporal data for past scene reconstruction. The investigation focused on the benefits of utilizing synthetic temporal data for the early detection of plant disease, specifically yellow rust in wheat, using aerial remote sensing imagery as the primary data source.

From the findings and discussions presented in Section V, the following conclusion were drawn.

1) The approach of representing plots as graphs, as opposed to relying solely on spectral profiles, shows promise in preserving both spatial and spectral information from the imagery.
2) Critical features extracted from imagery had the same importance in models trained with real data and with a combination of real and synthetic temporal data.
3) The integration of synthetic temporal data has the potential to enhance early plant disease detection by up to 53% compared with models trained exclusively on real data or balanced synthetic imagery.
4) The validation study revealed that the classification models trained with real imagery alone or a combination of real and synthetic temporal imagery utilized the same regions (i.e., pixels) of the control group.

Regarding the limitations, these primarily include the need for plot-specific temporal characterization and a comprehensive investigation into the tradeoff between spectral-based and graph-based approaches.

The achieved results open avenues to future work that might explore the opportunities of performing: 1) scene reconstruction utilizing the proposed method in other scenarios (e.g., animal trajectory and urban development); 2) data balancing; and 3) outlier detection in scarce temporal multispectral data contexts. In addition, it is worth considering the integration of diffusion-based approaches in these explorations. While the GAN formulation has been effective for our current problem representation, future work could investigate the potential benefits of leveraging diffusion models to enhance results and improve training stability.

Acknowledgment

The authors thank the members of the Remote Sensing Laboratory, Saint Louis University, St. Louis, MO, USA, for providing data collection, curation, and analytics support.

References

[1] C. Khosla and B. S. Saini, "Enhancing performance of deep learning models with different data augmentation techniques: A survey," in *Proc. Int. Conf. Intell. Eng. Manage. (ICIEM)*, Jun. 2020, pp. 79–85.
[2] J. Skobalski, V. Sagan, H. Alifu, O. Al Akkad, F. A. Lopes, and F. Grignola, "Bridging the gap between crop breeding and GeoAI: Soybean yield prediction from multispectral UAV images with transfer learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 210, pp. 260–281, Apr. 2024.
[3] S. Bhadra, V. Sagan, J. Skobalski, F. Grignola, S. Sarkar, and J. Vilbig, "End-to-end 3D CNN for plot-scale soybean yield prediction using multitemporal UAV-based RGB images," *Precis. Agricult.*, vol. 25, no. 2, pp. 834–864, Apr. 2024.
[4] P. Wu and J. Liu, "Learning causal temporal relation and feature discrimination for anomaly detection," *IEEE Trans. Image Process.*, vol. 30, pp. 3513–3527, 2021.
[5] Q. Zhang, Q. Yuan, C. Zeng, X. Li, and Y. Wei, "Missing data reconstruction in remote sensing image with a unified spatial–temporal–spectral deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4274–4288, Aug. 2018.
[6] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615.
[7] S. Berisha, F. F. Shahraki, D. Mayerich, and S. Prasad, "Deep learning for hyperspectral image analysis, Part I: Theory and algorithms," in *Hyperspectral Image Analysis* (Advances in Computer Vision and Pattern Recognition). Cham, Switzerland: Springer, 2020, pp. 37–68.
[8] J. Jagannathan and C. Divya, "Deep learning for the prediction and classification of land use and land cover changes using deep convolutional neural network," *Ecol. Inform.*, vol. 65, Nov. 2021, Art. no. 101412.
[9] C. Xu et al., "FastVSDF: An efficient spatiotemporal data fusion method for seamless data cube," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5402022.
[10] A. Azarang and N. Kehtarnavaz, "A generative model method for unsupervised multispectral image fusion in remote sensing," *Signal, Image Video Process.*, vol. 16, no. 1, pp. 63–71, Feb. 2022.
[11] M. Jiang, H. Shen, and J. Li, "Deep-learning-based spatio-temporal-spectral integrated fusion of heterogeneous remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5410915.
[12] C. Nguyen, V. Sagan, J. Skobalski, and J. I. Severo, "Early detection of wheat yellow rust disease and its impact on terminal yield with multi-spectral UAV-imagery," *Remote Sens.*, vol. 15, no. 13, p. 3301, Jun. 2023. [Online]. Available: https://www.mdpi.com/2072-4292/15/13/3301
[13] H. Shen et al., "Missing information reconstruction of remote sensing data: A technical review," *IEEE Geosci. Remote Sens. Mag.*, vol. 3, no. 3, pp. 61–85, Sep. 2015.
[14] N. Takeishi and A. Kalousis, "Physics-integrated variational autoencoders for robust and interpretable generative modeling," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 14809–14821.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:40:35 UTC from IEEE Xplore. Restrictions apply.

[15] F. Zhou, X. Liu, K. Zhang, and G. Trajcevski, "Toward discriminating and synthesizing motion traces using deep probabilistic generative models," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2401–2414, Jun. 2021.

[16] J. Yoon, D. Jarrett, and M. Van der Schaar, "Time-series generative adversarial networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–11.

[17] J. Jeon, J. Kim, H. Song, S. Cho, and N. Park, "GT-GAN: General purpose time series synthesis with generative adversarial networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 36999–37010.

[18] H. Peng, X. Chen, and J. Zhao, "Residual pixel attention network for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jun. 2020, pp. 486–487.

[19] R. Xu, M. Yao, C. Chen, L. Wang, and Z. Xiong, "Continuous spectral reconstruction from RGB images via implicit neural representation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)* (Lecture Notes in Computer Science), vol. 13805. Cham, Switzerland: Springer, 2022, pp. 78–94.

[20] J. Xiao, A. K. Aggarwal, U. K. Rage, V. Katiyar, and R. Avtar, "Deep learning-based spatiotemporal fusion of unmanned aerial vehicle and satellite reflectance images for crop monitoring," *IEEE Access*, vol. 11, pp. 85600–85614, 2023.

[21] H. Zhang, E. Fromont, S. Lefevre, and B. Avignon, "Multispectral fusion for object detection with cyclic fuse-and-refine blocks," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2020, pp. 276–280.

[22] C. M. Gevaert, J. Suomalainen, J. Tang, and L. Kooistra, "Generation of spectral–temporal response surfaces by combining multispectral satellite and hyperspectral UAV imagery for precision agriculture applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3140–3146, Jun. 2015.

[23] F. Yang et al., "Prediction of corn variety yield with attribute-missing data via graph neural network," *Comput. Electron. Agricult.*, vol. 211, Aug. 2023, Art. no. 108046.

[24] W. Wu, L. Ge, J. Luo, R. Huan, and Y. Yang, "A spectral–temporal patch-based missing area reconstruction for time-series images," *Remote Sens.*, vol. 10, no. 10, p. 1560, 2018.

[25] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, p. 111716, May 2020.

[26] D. A. Kasampalis, T. K. Alexandridis, C. Deva, A. Challinor, D. Moshou, and G. Zalidis, "Contribution of remote sensing on crop models: A review," *J. Imag.*, vol. 4, no. 4, p. 52, 2018.

[27] O. Amodu, R. Nordin, C. Jarray, U. Bukar, R. R. Mahmood, and M. Othman, "A survey on the design aspects and opportunities in age-aware UAV-aided data collection for sensor networks and Internet of Things applications," *Drones*, vol. 7, no. 4, p. 260, Apr. 2023.

[28] M.-F. Racault, S. Sathyendranath, and T. Platt, "Impact of missing data on the estimation of ecological indicators from satellite ocean-colour time-series," *Remote Sens. Environ.*, vol. 152, pp. 15–28, Sep. 2014.

[29] D. N. H. Horler, M. Dockray, and J. Barber, "The red edge of plant leaf reflectance," *Int. J. Remote Sens.*, vol. 4, no. 2, pp. 273–288, Jan. 1983.

[30] J. Peñuelas and I. Filella, "Visible and near-infrared reflectance techniques for diagnosing plant physiological status," *Trends Plant Sci.*, vol. 3, no. 4, pp. 151–156, 1998.

[31] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," 2015, *arXiv:1511.06434*.

[32] J. Diaz, D. Mitsche, G. Perarnau, and X. Pérez-Giménez, "On the relation between graph distance and Euclidean distance in random geometric graphs," *Adv. Appl. Probab.*, vol. 48, no. 3, pp. 848–864, 2016.

[33] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[34] B. Zhou, "Image segmentation using slic superpixels and affinity propagation clustering," *Int. J. Sci. Res*, vol. 4, no. 4, pp. 1525–1529, 2015.

[35] T. Kattenborn, J. Leitloff, F. Schiefer, and S. Hinz, "Review on convolutional neural networks (CNN) in vegetation remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 24–49, Mar. 2021.

[36] F. A. Lopes, V. Sagan, and F. Esposito, "PlantPlotGAN: A physics-informed generative adversarial network for plant disease prediction," 2023, *arXiv:2310.18268*.

[37] J. W. Rouse et al., "Monitoring vegetation systems in the great plains with ERTS," *NASA Special Publications*, vol. 351, no. 1, p. 309, 1974.

[38] A. A. Gitelson and M. N. Merzlyak, "Remote sensing of chlorophyll concentration in higher plant leaves," *Adv. Space Res.*, vol. 22, no. 5, pp. 689–692, Jan. 1998.

[39] G. S. Birth and G. R. McVey, "Measuring the color of growing turf with a reflectance spectrophotometer," *Agronomy J.*, vol. 60, no. 6, pp. 640–643, 1968.

[40] A. A. Gitelson, Y. Gritz, and M. N. Merzlyak, "Relationships between leaf chlorophyll content and spectral reflectance and algorithms for non-destructive chlorophyll assessment in higher plant leaves," *J. Plant Physiol.*, vol. 160, no. 3, pp. 271–282, Jan. 2003.

[41] C. F. Jordan, "Derivation of leaf-area index from quality of light on the forest floor," *Ecology*, vol. 50, no. 4, pp. 663–666, Jul. 1969.

[42] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, May 1979.

[43] A. R. Huete, "A soil-adjusted vegetation index (SAVI)," *Remote Sens. Environ.*, vol. 25, no. 3, pp. 295–309, Aug. 1988.

[44] J. Qi, A. Chehbouni, A. R. Huete, Y. H. Kerr, and S. Sorooshian, "A modified soil adjusted vegetation index," *Remote Sens. Environ.*, vol. 48, no. 2, pp. 119–126, 1994.

[45] G. Rondeaux, M. Steven, and F. Baret, "Optimization of soil-adjusted vegetation indices," *Remote Sens. Environ.*, vol. 55, no. 2, pp. 95–107, Feb. 1996.

[46] J.-L. Roujean and F.-M. Breon, "Estimating PAR absorbed by vegetation from bidirectional reflectance measurements," *Remote Sens. Environ.*, vol. 51, no. 3, pp. 375–384, Mar. 1995.

[47] N. H. Broge and E. Leblanc, "Comparing prediction power and stability of broadband and hyperspectral vegetation indices for estimation of green leaf area index and canopy chlorophyll density," *Remote Sens. Environ.*, vol. 76, no. 2, pp. 156–172, 2001.

[48] F. Baret, G. Guyot, and D. Major, "Crop biomass evaluation using radiometric measurements," *Photogrammetria*, vol. 43, no. 5, pp. 241–256, 1989.

[49] A. J. Richardson and C. L. Wiegand, "Distinguishing vegetation from soil background information," *Photogramm. Eng. Remote Sens.*, vol. 43, no. 12, pp. 1541–1552, 1977.

[50] D. Major, F. Baret, and G. Guyot, "A ratio vegetation index adjusted for soil brightness," *Int. J. Remote Sens.*, vol. 11, no. 5, pp. 727–740, 1990.

[51] F. Baret and G. Guyot, "Potentials and limits of vegetation indices for LAI and APAR assessment," *Remote Sens. Environ.*, vol. 35, nos. 2–3, pp. 161–173, Feb. 1991.

[52] S. K. McFeeters, "The use of the normalized difference water index (NDWI) in the delineation of open water features," *Int. J. Remote Sens.*, vol. 17, no. 7, pp. 1425–1432, May 1996.

[53] J. Peñuelas, J. Gamon, A. Fredeen, J. Merino, and C. Field, "Reflectance indices associated with physiological changes in nitrogen- and water-limited sunflower leaves," *Remote Sens. Environ.*, vol. 48, no. 2, pp. 135–146, 1994.

[54] M. Shibayama and T. Akiyama, "Seasonal visible, near-infrared and mid-infrared spectra of Rice canopies in relation to LAI and above-ground dry phytomass," *Remote Sens. Environ.*, vol. 27, no. 2, pp. 119–127, Feb. 1989.

[55] C. S. T. Daughtry, C. L. Walthall, M. S. Kim, E. B. de Colstoun, and J. E. McMurtrey, "Estimating corn leaf chlorophyll concentration from leaf and canopy reflectance," *Remote Sens. Environ.*, vol. 74, no. 2, pp. 229–239, 2000.

[56] D. Haboudane, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, Apr. 2004. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425704000264

[57] J. U. H. Eitel, D. S. Long, P. E. Gessler, and A. M. S. Smith, "Using in-situ measurements to evaluate the new RapidEye satellite series for prediction of wheat nitrogen status," *Int. J. Remote Sens.*, vol. 28, no. 18, pp. 4183–4190, Sep. 2007, doi: 10.1080/01431160701422213.

[58] A. R. Huete, K. Didan, T. Miura, E. P. Rodriguez, X. Gao, and L. G. Ferreira, "Overview of the radiometric and biophysical performance of the MODIS vegetation indices," *Remote Sens. Environ.*, vol. 83, nos. 1–2, pp. 195–213, 2002. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425702000962

[59] B. Datt, "A new reflectance index for remote sensing of chlorophyll content in higher plants: Tests using eucalyptus leaves," *J. Plant Physiol.*, vol. 154, no. 1, pp. 30–36, Jan. 1999. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0176161799803149

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[60] A. Marshak, Y. Knyazikhin, A. Davis, W. Wiscombe, and P. Pilewskie, "Cloud-vegetation interaction: Use of normalized difference cloud index for estimation of cloud optical thickness," *Geophys. Res. Lett.*, vol. 27, no. 12, pp. 1695–1698, 2000.

[61] M. N. Merzlyak and A. E. Solovchenko, "Photostability of pigments in ripening apple fruit: A possible photoprotective role of carotenoids during plant senescence," *Plant Sci.*, vol. 163, no. 4, pp. 881–888, Oct. 2002.

[62] J. Penuelas, F. Baret, and I. Filella, "Semi-empirical indices to assess carotenoids/chlorophyll a ratio from leaf spectral reflectance," *Photosynthetica*, vol. 31, no. 2, pp. 221–230, 1995.

[63] M. Vincini, E. Frazzi, and P. D'Alessio, "Angular dependence of maize and sugar beet VIs from directional CHRIS/Proba data," in *Proc. 4th ESA CHRIS PROBA Workshop*, 2006, pp. 19–21.

[64] D. Haboudane, J. R. Miller, N. Tremblay, P. J. Zarco-Tejada, and L. Dextraze, "Integrated narrow-band vegetation indices for prediction of crop chlorophyll content for application to precision agriculture," *Remote Sens. Environ.*, vol. 81, nos. 2–3, pp. 416–426, Aug. 2002.

[65] M. Xu et al., "Retrieving leaf chlorophyll content using a matrix-based vegetation index combination approach," *Remote Sens. Environ.*, vol. 224, pp. 60–73, Apr. 2019.

[66] E. Barnes et al., "Coincident detection of crop water stress, nitrogen status and canopy density using ground based multispectral data," in *Proc. 5th Int. Conf. Precis. Agricult.*, Bloomington, MN, USA, 2000, vol. 1619, no. 6.

[67] J. Dash and P. Curran, *Int. J. Remote Sens.*, vol. 25, no. 23, pp. 5403–5413, 2000.

[68] Z. Jiang, A. Huete, K. Didan, and T. Miura, "Development of a two-band enhanced vegetation index without a blue band," *Remote Sens. Environ.*, vol. 112, no. 10, pp. 3833–3845, Oct. 2008.

[69] D. M. Woebbecke, G. E. Meyer, K. Von Bargen, and D. A. Mortensen, "Color indices for weed identification under various soil, residue, and lighting conditions," *Trans. ASAE*, vol. 38, no. 1, pp. 259–269, 1995.

[70] C. J. Strong, N. G. Burnside, and D. Llewellyn, "The potential of small-unmanned aircraft systems for the rapid detection of threatened unimproved grassland communities using an enhanced normalized difference vegetation index," *PLoS ONE*, vol. 12, no. 10, Oct. 2017, Art. no. e0186193.

[71] A. A. Gitelson, M. N. Merzlyak, and O. B. Chivkunova, "Optical properties and nondestructive estimation of anthocyanin content in plant leaves¶," *Photochem. Photobiol.*, vol. 74, no. 1, pp. 38–45, Jul. 2001.

[72] A. A. Gitelson, Y. Zur, O. B. Chivkunova, and M. N. Merzlyak, "Assessing carotenoid content in plant leaves with reflectance spectroscopy¶," *Photochem. Photobiol.*, vol. 75, no. 3, pp. 272–281, 2002.

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